STAFF REVIEW · H446 083 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Updates a tentative distance via a cheaper path.

2. Connection with a cost.

3. Non-negative edge weights.

4. A,B,C,D.

I Do — source worked steps

1. Update neighbours — Start with dist(P)=0 . Settle P (smallest unsettled value). Relax P→Q: Q=3 , predecessor Q←P. Relax P→R: R=8 , predecessor R←P. Remaining unsettled distances: Q=3, R=8.

2. Improve the tentative route — Q=3 is the smallest unsettled distance. Via Q→R, the candidate is 3+2=5 . This beats R=8, so update R=5 and predecessor R←Q. Settled now: P,Q.

3. Reconstruct the actual path — R=5 is next, and is the target. Follow predecessor pointers R←Q←P ; reverse to obtain P→Q→R , total cost 5 . No vertex with a larger tentative distance was prematurely finalised.

We Do — guided CHECK

1. Settle X; calculate both tentative distances
Start at X=0. Which neighbour gets distance 4? Which gets 10? Which unsettled vertex should Dijkstra select next?
Settle Y and relax Y→Z
After X: Y=4 (predecessor X), Z=10 (predecessor X). Settle Y=4; candidate cost to Z is 4+3=7 , so improve Z to 7 with predecessor Y.
Finish with the path, not just the cost
Settle Z=7. Predecessors: Z←Y←X . Reverse to X→Y→Z , total cost 7. The direct X→Z cost 10 is longer.

You Do — source mark schemes

1. After A B2,C5,D∞ [3]; after B C3,D6 [3].

2. C dist3 [1], relax C→D cost1 gives D4 [2], predecessors D←C←B←A [2], path cost4 [1].

3. Finalise an apparently shortest vertex [2], later negative edge can improve it [2], so settled assumption false [1], use suitable negative-weight method instead [1].

Exit Ticket — source answer

1. False. The standard algorithm assumes non-negative weights.